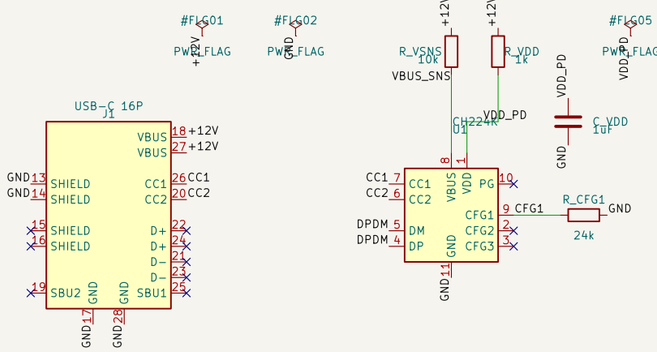
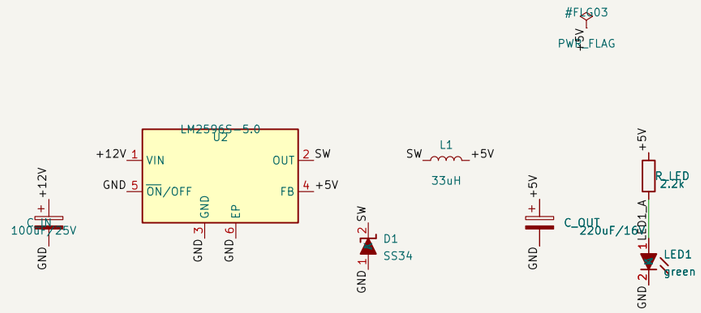
1. USB-C PD sink (CH224K -> 12 V)

CFG1 = 24k -> 12 V | VDD via 1k + 1uF | VBUS-sense 10k | DP/DM kortgesloten (PD-only)



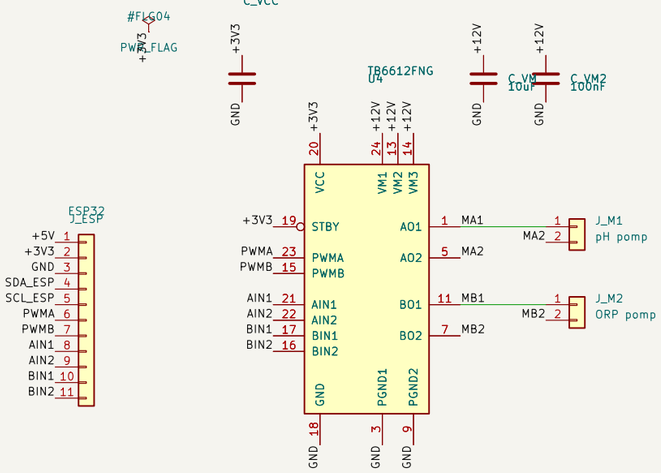
2. Buck 12 V -> 5 V (LM2596S-5.0)

SS34 + 33uH CD54 | 100uF/25V in, 220uF/16V uit | LED1 = 5 V aanwezig



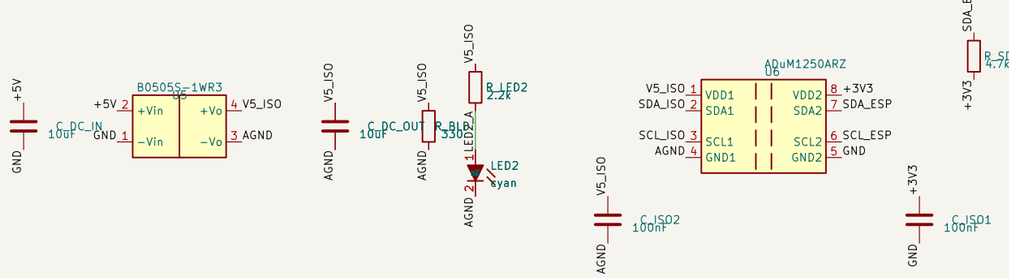
3. Motordriver TB6612FNG + ESP32-header

VCC/STBY op 3V3 van de ESP | VM op 12 V | 2x pomp-header



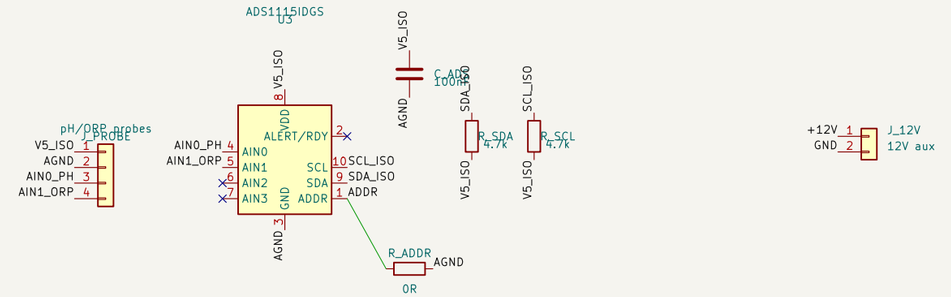
4. Isolatiebarriere (B0505S + ADuM1250)

side 1 = iso (V5_ISO/AGND), side 2 = ESP @3V3 | R_BLD = minimumlast | pull-ups aan beide kanten



5. Geisoleerde ADC (ADS1115)

ADDR = OR -> AGND (0x48) | pH op AIN0, ORP op AIN1 | probe-header op V5_ISO/AGND



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File: pura-mainboard-0.4.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
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